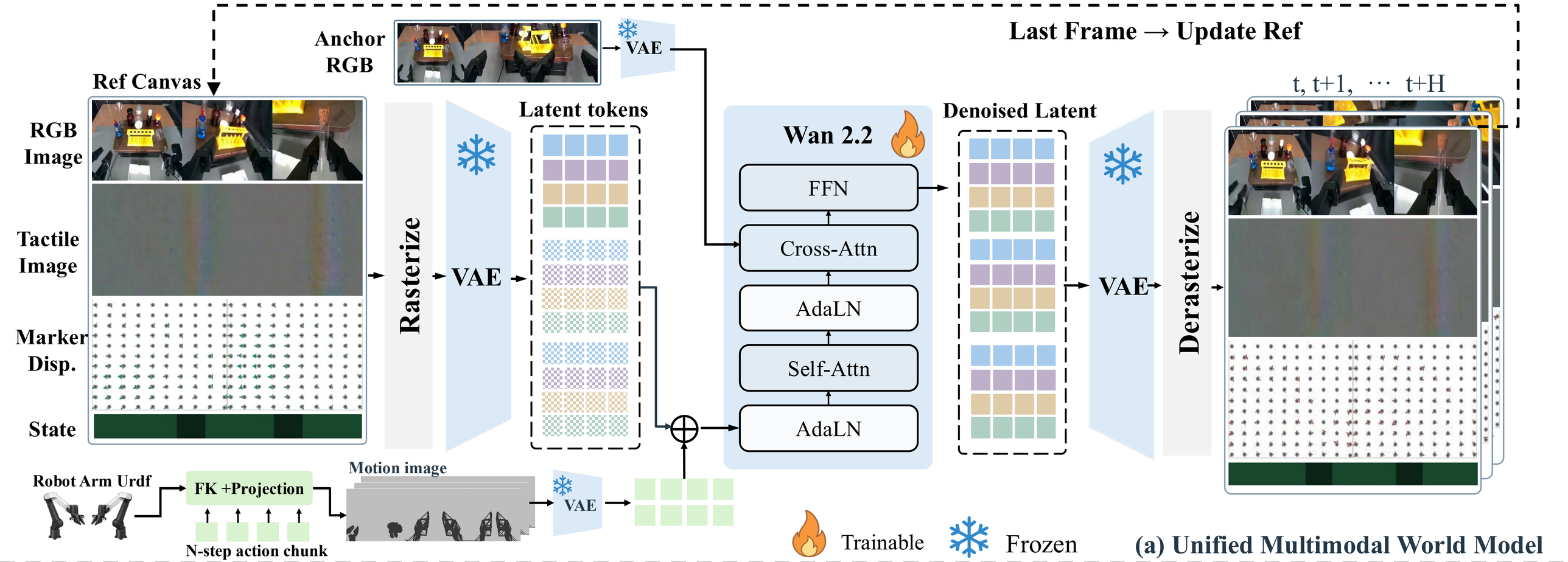
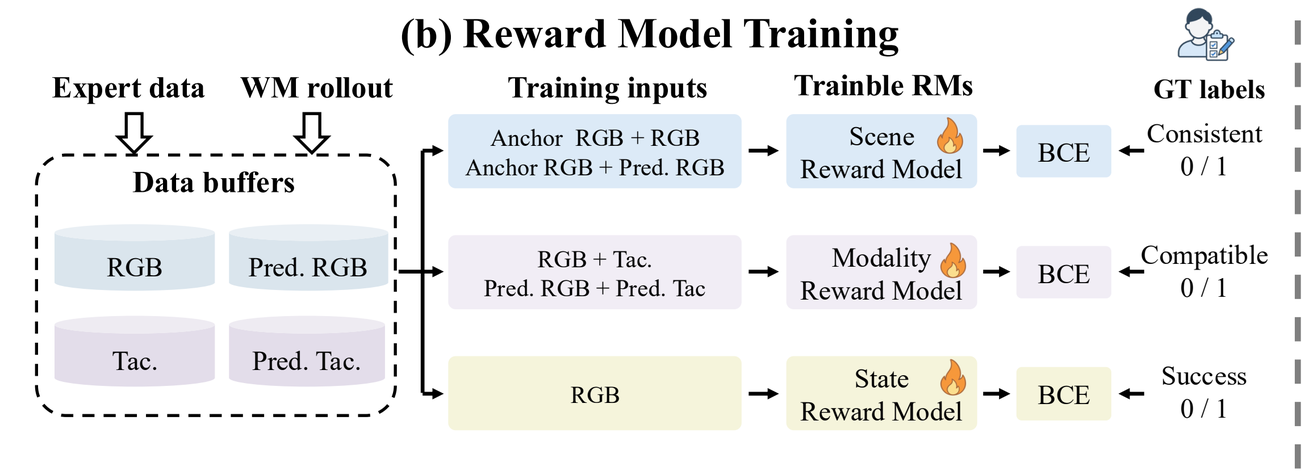
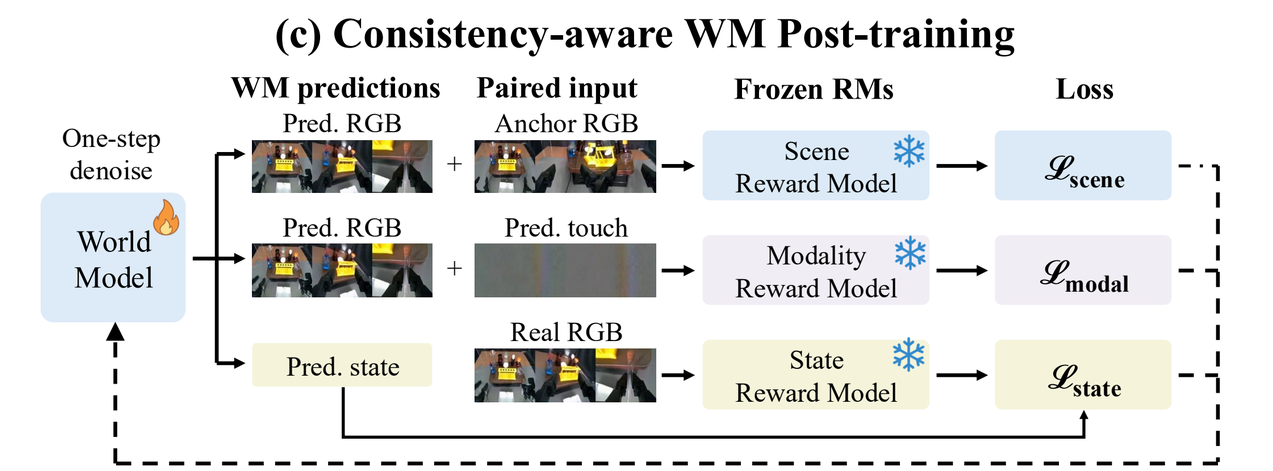

Last Frame → Update Ref
Anchor RGB
VAE
Ref Canvas
RGB Image
Tactile Image
Marker Disp.
State
Rasterize
VAE
Latent tokens
Wan 2.2
FFN
Cross-Attn
AdaLN
Self-Attn
AdaLN
Denoised Latent
VAE
Derasterize
t, t+1, ⋯ t+H
Robot Arm Urdf
FK +Projection
N-step action chunk
Motion image
VAE
Trainable
Frozen
(a) Unified Multimodal World Model


(b) Reward Model Training
Expert data
WM rollout
Data buffers
RGB
Pred. RGB
Tac.
Pred. Tac.
Training inputs
Trainble RMs
GT labels
Anchor RGB + RGB
Anchor RGB + Pred. RGB
Scene Reward Model
BCE
Consistent 0 / 1
RGB + Tac.
Pred. RGB + Pred. Tac
Modality Reward Model
BCE
Compatible 0 / 1
RGB
State Reward Model
BCE
Success 0 / 1


(c) Consistency-aware WM Post-training
One-step denoise
World Model
WM predictions
Paired input
Frozen RMs
Loss
Pred. RGB
Anchor RGB
Scene Reward Model
$\mathscr{L}_{scene}$
Pred. RGB
Pred. touch
Modality Reward Model
$\mathscr{L}_{modal}$
Pred. state
Real RGB
State Reward Model
$\mathscr{L}_{state}$